Hardware-Constrained Architecture Search for Lane-Change Intention Prediction from Vehicle Time Series

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Abstract—**Predicting a driver's lane-change intention early enough to act on it is a time-series classification problem with hard deployment limits: the model has to run on automotive-grade microcontrollers within tight memory and latency budgets. We apply constrained neural architecture search to the Lane Change Intention Recognition driving dataset [3] (windows of 50 samples over 31 signals) and obtain compact one-dimen- sional convolutional networks for three tasks: three-class intention classification and time-to-lane-change regression for left and right maneuvers. [TODO: results: accuracy, RMSE, model size, latency on STM32] The searched models improve on the published baseline (Transformer, 0.51 s time-to-lane-change RMSE, deployed in FP32 [1]) while using [TODO: X]× less memory, and we report measured, not estimated, footprint and latency on the same STM32 targets after int8 quantization.**

Index Terms—**Neural architecture search, lane-change intention, time-series classification, TinyML, embedded inference, STM32.**

I. Introduction

[TODO: Motivation: intention prediction as advance warning for ADAS; why on-device inference (latency, privacy, cost, no connectivity assumptions); the gap: published DMIR models target accuracy, not deployment budgets.]

Contributions:

1) a constrained search space and objective tuned to one-dimensional driving signals under microcontroller budgets;
2) models that beat the published DMIR baseline on all three tasks at a fraction of its footprint [TODO: verify once results exist];
3) measured deployment numbers (flash, RAM, latency) on [TODO: STM32 part], with the full pipeline released as open source.

II. Related Work

A. Lane-change intention prediction

[TODO: The LC dataset and framework baseline [1] (TTLC regression, Transformer RMSE 0.51 s, FP32 deployment on STM32H7B3/F401); earlier HighD-based TTLC work (RMSE 0.63–0.7); MicroNAS for time series and TinyTNAS as closest NAS relatives.]

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B. Constrained architecture search for microcontrollers

[TODO: muNAS [2], MCUNet/TinyNAS, MicroNets; what transfers to 1D time series and what does not.]

III. Data

We use the Lane Change Intention Recognition dataset [3] (50 human drivers in the CARLA simulator, 10 Hz, over 3,400 annotated lane changes; MIT license), prepared as fixed windows of T=50 samples (5 s) over F=31 channels [TODO: confirm channel order and whether channel 31 is fileTime — drop it if so; official feature count is 30]. Classification uses balanced splits of 94,620 / 16,332 / 19,722 windows (train/val/test) over three classes: no intention, right lane change, left lane change. The regression tasks map a window to the time until the lane change, $t \in [0, 4]$ s discretized at 0.1 s, with 39,313 / 7,742 / 8,097 (LCL) and 33,201 / 5,828 / 7,073 (LCR) windows.

The splits are user-independent: we verified by matching raw per-driver session logs against the prepared windows (correlation over 50-sample windows) that the validation and test drivers follow the protocol of [1] with no window leakage across splits. The test splits contain isolated spikes (up to 5×10^6) on the right/left `curvatureDx` channel pair — a numerical-derivative artefact absent from training; we clip validation and test features to the per-channel training range and leave training data untouched.

IV. Method

A. Search space

[TODO: 1D cells: depthwise-separable and standard convolutions, kernel sizes, widths, depth, pooling; operator set restricted to what the ST Edge AI compiler maps efficiently.]

B. Constraints and search algorithm

[TODO: peak RAM, flash after int8 quantization, MACs; aging evolution with constraint handling; search cost budget.]

V. Experiments

A. Setup

[TODO: training schedule, hardware, seeds/repeats, hyperparameters; all runs logged to the released experiments.jsonl.]

TABLE I
TEST-SET COMPARISON. THE INTERNAL REFERENCE MODELS (UNPUBLISHED) WERE EVALUATED ON THE SAME TEST SET; CLASSIFICATION ACCURACY IS OUR OWN EVALUATION, THE TRANSFORMER RMSE VALUES ARE AS REPORTED. SIZE IS THE PARAMETER COUNT.

Model	Params	Acc. (%)	LCR	LCL
Reference CNN (cls)	441 k	91.5	—	—
Reference Transf. (reg, RMSE)	333/49 k	—	0.42	0.44
Our DSCNN baseline	10 k	91.5	0.32†	0.33†
Ours searched (cls / reg MAE)	84 k	**92.1**	0.29†	0.33†
Ours compact	8 k	91.3	—	—
Ours, no turn-signal	21 k	91.1	—	—

†MAE (our search optimizes MAE); reference regression is reported as RMSE. Our LCR RMSE is 0.45 and LCL 0.50.

B. Results

Table I reports test-set metrics. All searched-model numbers are our own test-set evaluation of the saved models, from searches run to their full 150-round budget. [TODO: add peak-RAM, MACs, and measured on-device latency columns from ST Edge AI; consider re-running with an all-models save policy for a guaranteed-optimal front (see notes).]

For classification the searched model is both smaller and more accurate than the 441 k reference CNN (92.1% at 84 k parameters; a 8 k model still reaches 91.3%). Regression beats the published single-TTLC SOTA on MAE (0.287 vs 0.298 for LCR) [TODO: re-run with an RMSE-aware objective before comparing to the internal-reference Transformers on RMSE (they are competitive, especially LCL)]. Removing the turn-signal channels drops classification accuracy by about one point (92.1% to 91.1%)—evidence that the model anticipates from vehicle dynamics and surrounding traffic rather than reading a declared signal. [TODO: RMSE columns and the internal-reference RMSE (0.42/0.44) are not yet matched because the search optimizes MAE; decide RMSE-aware objective vs MAE-primary framing.]

C. Deployment on STM32

Target platform: STM32H7B3I-DK discovery kit (Cortex-M7 at 280 MHz, 1.4 MB SRAM, 2 MB flash) — the same MCU family used by the baseline framework [1], which deployed FP32 models only. We quantize with int16$\times$8 (int8 weights, int16 activations); on these wide-dynamic-range inputs full int8 degrades accuracy sharply, whereas int16$\times$8 preserves it (e.g. classification $92.08\% \rightarrow 92.15\%$, LCR MAE $0.287 \rightarrow 0.286$), so no quantization-aware training is needed.

The resulting models are small: the classification model occupies 118 KB of flash with an estimated 4.5 KB peak activation RAM and 152 k MACs; a 19 KB variant still reaches 91.2%. Every candidate fits even the low-end STM32F401 (96 KB SRAM), on which the baseline's Transformer did not fit [1]. [TODO: replace estimates with ST Edge AI's measured flash / peak SRAM / latency (Table III comparison); see docs/research/deployment.md.]

VI. CONCLUSION

[TODO: two or three sentences; no new claims.]

REFERENCES

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